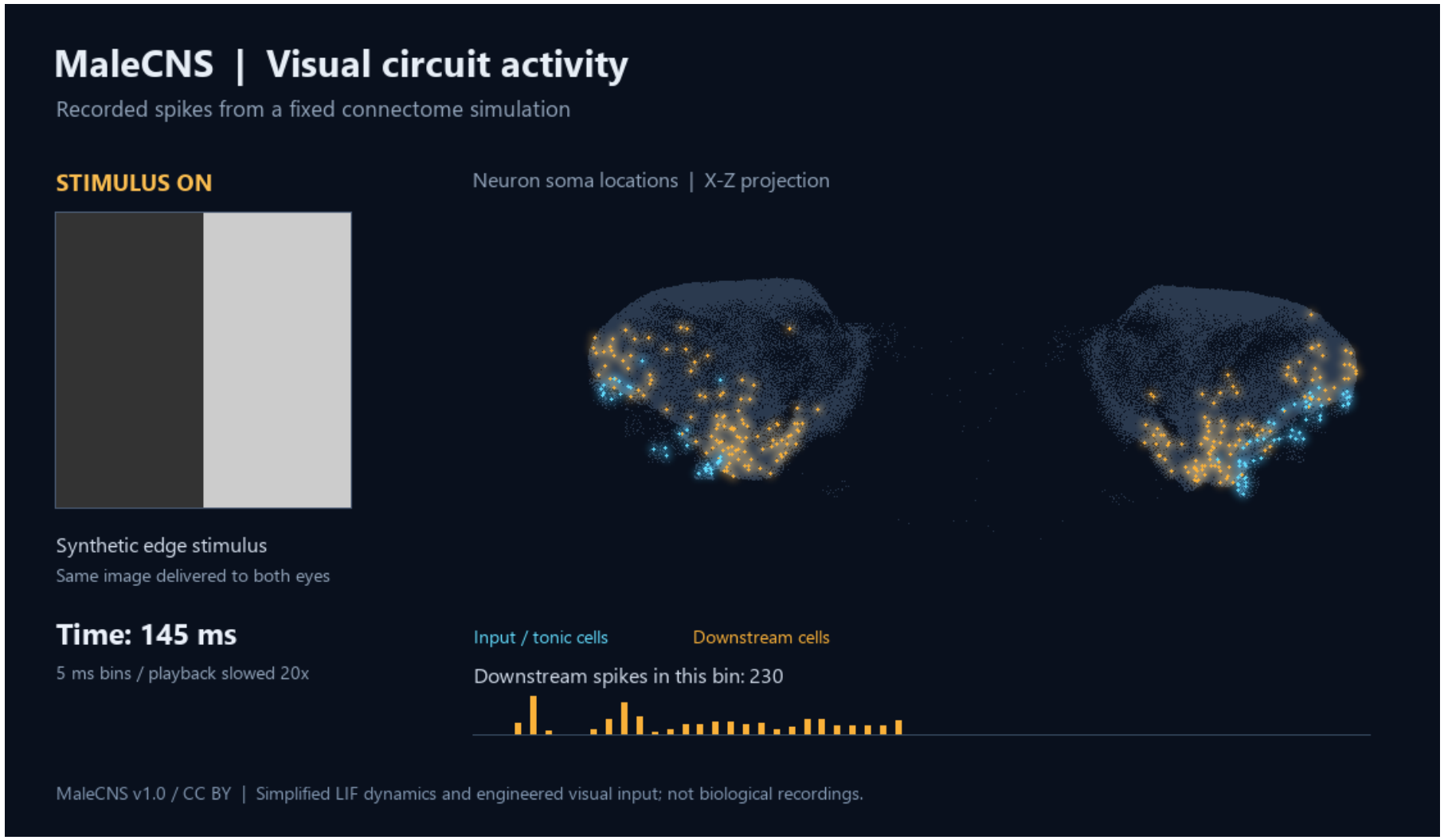
A small connectome experiment

For a bit of fun: bacterial versus fungal keratitis

Fixed fly circuitry, simulated responses, and a small supervised classifier.



Synthetic input and simulated activity. No patient photographs.

How the experiment worked

1 Image to stimulation

A 128 by 128 grayscale image was mapped to visual input neurons. Both eyes received the same image.

2 A fixed visual circuit

A simplified LIF simulation used 101,939 neurons and 11.76 million directed connections from MaleCNS.

3 Response features

After a gray baseline, the image was presented for 100 ms. Readout spikes were counted in two 50 ms windows.

4 A small learned readout

Per-image log1p counts were averaged within visits. Scaling, PCA to 28 dimensions, and logistic regression used training patients only.

5 Patient-separated comparison

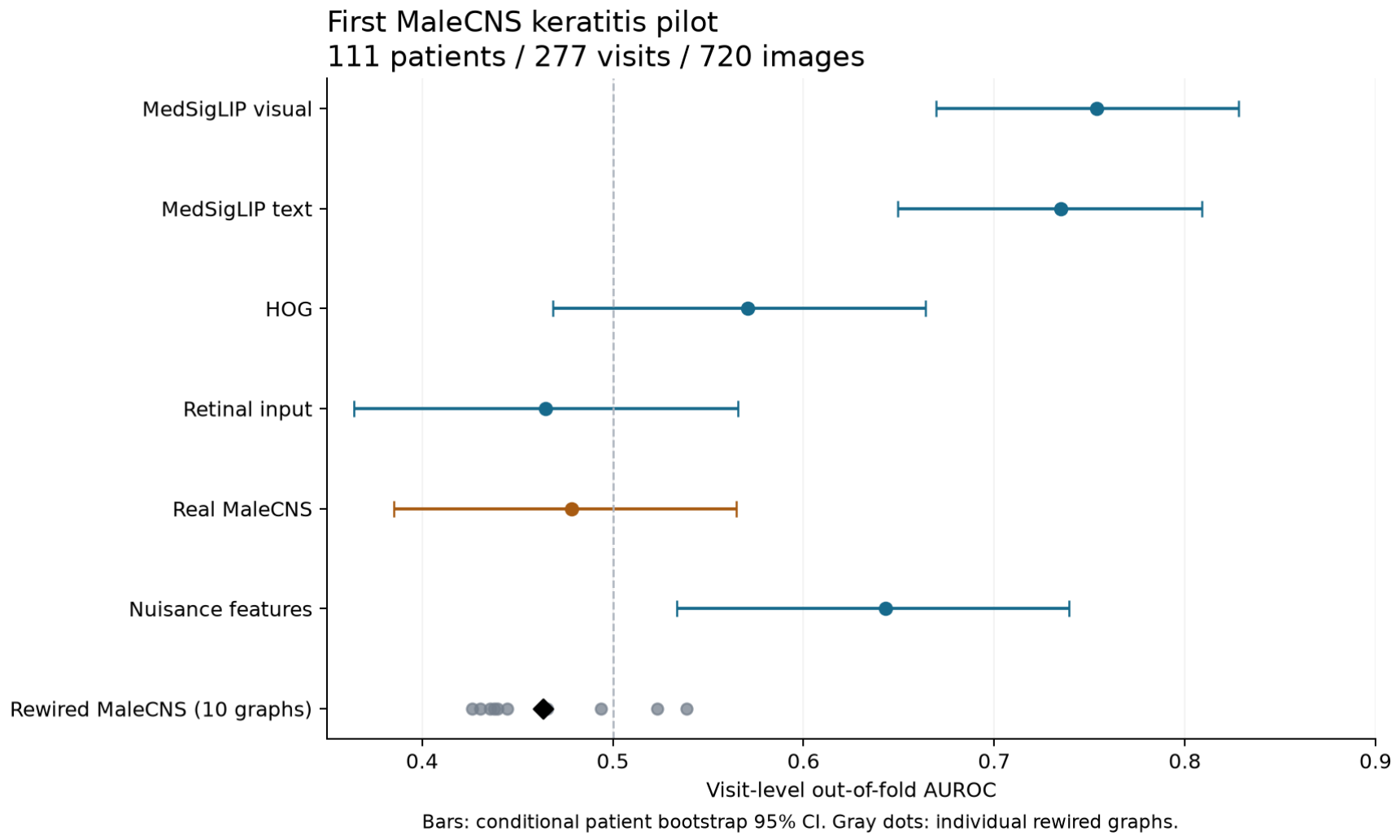
Five outer folds kept each patient's images and visits together. Real wiring was compared with raw input and ten rewired circuits.

The circuit stayed frozen. Only the final readout learned from labels.

MaleCNS: Janelia/FlyEM, Cambridge, MRC LMB, Google Research and collaborators (CC BY 4.0)
male-cns.janelia.org/download/ | Simulation implementation adapted through Fly OCR (MIT)

The results were around chance

111 patients / 277 visits / 720 images



0.48

Visit AUROC
Masked crop

0.46

Visit AUROC
Whole image

No convincing advantage over rewired controls in this setup.

Crop real minus rewired mean: +0.015 AUROC (paired 95% interval -0.037 to +0.064).

These are internal cross-validation results for one input, simulation, and readout setup. The intervals condition on fixed OOF predictions; there was no external validation.

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